Crisscross-Global Vision Transformers Model for Very High Resolution Aerial Image Semantic Segmentation

Guohui Deng, Zhaocong Wu, Miaozhong Xu, Chengjun Wang, Zhiye Wang, and Zhongyuan Lu

***Abstract*— Semantic segmentation is a key means for understanding very high resolution (VHR) aerial imagery. With the explosive development of deep learning, deep learning methods are being applied to the segmentation of VHR images, with convolutional neural networks (CNNs) as the basic framework; however, owing to the highly complex details present in VHR images and the high spatial dependence of geographical objects, CNN-based methods are inadequate. This is because the inherent locality of CNNs limits the size of the receptive field, thus limiting the ability to obtain long-range context information. To solve this problem, in this article, we propose a transformer-based novel deep learning model called crisscross-global vision transformers (CGVTs). CGVT exploits the transformer's inherent ability to obtain long-range context information to solve the restricted receptive field problem. Specifically, we redesign the self-attention (SA) mechanism in the transformer and call it crisscross-global attention. It consists of two parts: a crisscross transformer encoder block (CC-TEB) and a global squeeze transformer encoder block (GS-TEB). CC-TEB overcomes the limitation of the traditional SA design (specifically, difficulty applying it to VHR aerial image segmentation) and further increases the local feature representation ability of the model. GS-TEB increases the global feature representation ability of the model. The results of experiments conducted on the popular ISPRS Vaihingen, IEEE GRSS data fusion contest Zeebrugge, and LoveDA semantic segmentation challenge datasets verify the effectiveness and superiority of our proposed method. Specifically, it achieved state-of-the-art performance on both Zeebrugge and LoveDA datasets and is currently ranked second in the Vaihingen dataset.**

***Index Terms*— Aerial imagery, remote sensing, semantic segmentation, transformers.**

Manuscript received 10 March 2022; revised 27 July 2022, 28 November 2022, and 23 March 2023; accepted 18 April 2023. Date of publication 15 May 2023; date of current version 1 June 2023. This work was supported by the National Key Research and Development Program of China under Grant 2018YFB0504800 and Grant 2018YFB0504801. *(Corresponding authors: Zhaocong Wu; Miaozhong Xu.)*

Guohui Deng is with the School of Internet, Anhui University, Anhui 230049, China, and also with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: dlyj0207@whu.edu.cn).

Zhaocong Wu is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: zcwoo@whu.edu.cn).

Miaozhong Xu, Chengjun Wang, Zhiye Wang, and Zhongyuan Lu are with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: mzxu6319@whu.edu.cn; ChengjunWang@whu.edu.cn; zhiye.wang@whu.edu.cn; Luzhongyuan@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3276172

I. Introduction

VERY-HIGH resolution (VHR) aerial images are widely used in activities such as land resources monitoring and illegal building monitoring because of their high spatial resolution and high timeliness. The key when using VHR aerial images is ensuring that accurate information is extracted. In image semantic segmentation, each pixel in the image is assigned to a unique semantic category. Therefore, it is necessary to study the semantic segmentation of VHR aerial images.

At present, with the development of deep learning technology, semantic segmentation for VHR aerial images has advanced from traditional methods based on boundary, region, or Markov random field [1], [2] to deep learning-based methods. Most semantic segmentation methods based on deep learning use the encoder–decoder structure [3], [4]. The encoder contains numerous stacked convolution layers, each with a specific kernel size. The convolution layers perform translation traversal on the processed image to obtain the feature representation. A certain convolution kernel traverses the whole image, and the matrix parameters of the convolution kernel are fixed, so the convolution has a unique advantage: shared parameters and translation invariance. The decoder performs pixel-wise classification of the representation features obtained from the encoder.

For image data, long-range context information reflects the connection between the positions of entities that are far apart in an image. For VHR aerial images, various geographical elements have a higher dependence on geographical space. For example, roads are usually close to buildings, and cars are mostly located on roads. The distribution of objects is often very scattered, especially some fine-grained objects, such as cars and boats, and these kinds of fine-grained objects also have strong dependencies. Consequently, capturing long-range dependencies is of central importance in deep neural networks [5]. It is, however, difficult to obtain long-range context information using convolutional neural networks (CNNs) owing to their inherent locality.

Whether in the field of CV or remote sensing, in semantic segmentation tasks, all CNN-based methods [6], [7], [8] try to obtain long-range context information to improve the accuracy of segmentation. The most common method is to increase the size of the receptive field, such as using the dilated convolution strategy. It is, however, impossible to increase the size of the

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

receptive field indefinitely because doing so increases the computation amount and continuity information is lost, resulting in missing object details. Furthermore, although a deeper CNN theoretically should have a large receptive field [9], studies have reported the converse [10].

Recently, a new deep learning method called transformer [11] has garnered interest in the natural language processing (NLP) arena and has become a state-of-the-art approach. The process employed by the transformer differs from that used in CNNs; it is a sequence-to-sequence structure composed of self-attention (SA) layers without the convolution operation. The feature extraction ability of the transformer and its ability to obtain long-range context information significantly exceed that of CNN-based methods and are critical for semantic segmentation tasks. Consequently, many CV researchers have applied the transformer directly to image classification, with the most successful method being ViT [12].

ViT is also based on the encoder–decoder structure. The encoder of ViT consists of numerous SA layers. In other words, the transformer extracts the features through SA rather than convolution. This serialized SA mechanism enables the transformer to extract long-range context information. Naturally, we apply the transformer to the segmentation of remote sensing images. To the best of our knowledge, only a few studies have applied the transformer to the semantic segmentation of VHR aerial images.

Although ViT successfully applied the transformer to image classification, it still has problems with pixel-wise dense prediction, such as semantic segmentation. The main reasons are as follows. First, because there are numerous SA operations in the transformer, there are numerous parameters, which makes training difficult. Second, the feature map output by the transformer encoder is single-scale and has a small resolution. Third, learnable position encoding is used in ViT [12], which is a vector with the same length as the input sequence. Once this vector is trained, the position encoding is fixed and, during testing, once the size of the input is different from the training input case, the performance decreases. This destroys the translation invariance of the model, which significantly affects the generalization ability of the model.

To solve the above problems, we restructured the deep network with crisscross-global vision transformers (CGVTs) for VHR aerial image semantic segmentation. The crisscross-global vision transformer (ViT) is an encoder structure with a transformer as the backbone. Specifically, first, the input image is divided into several nonoverlapping image patches through the image partition operation. We regard these patches as "tokens," similar to ViT. These tokens are then projected into any dimension using an embedding layer, which is followed by position encoding of the high-level features using conditional position encoding (CPE). Then, the obtained feature map is fed into our redesigned transformer block.

The redesigned transformer block differs from that of ViT in the following aspects. 1) inspired by the network structure design of CNNs, we design a pyramid-like progressively reduced transformer encoder block, which aims to reduce the amount of computation while obtaining scale features; 2) to further reduce the amount of computation, we perform squeeze scaling on key (K), multivalue (V), called the "attention squeeze (AS) strategy," to further reduce the amount of computation when performing SA; and 3) meanwhile, inspired by the network structure of [12], [13], [14], and [15] we redesigned the SA structure to obtain crisscross-global attention. It consists of two parts: a crisscross transformer encoder block (CC-TEB) and a global squeeze transformer encoder block (GS-TEB). CC-TEB overcomes the excessive calculation in the traditional ViT design while further increasing the model's local feature representation. GS-TEB increases the global representation power of the model. Finally, the result obtained by the crisscross-global transformer block is sent to the decoder for decoding to obtain the segmentation result.

The main contributions of this study are as follows.

1) We designed a novel end-to-end deep network called CGVTs for VHR aerial image semantic segmentation. CGVT is a novel encoder structure based on the transformer that fully uses the transformer's ability to obtain long-distance context information and successfully applies it to remote sensing image segmentation.
2) We propose CC-TEB and GS-TEB to achieve the difficult balance between the high computational cost and the local–global information acquisition in the application of the transformer to remote sensing image segmentation.
3) The proposed CGVT has a strong performance on VHR aerial image semantic segmentation. We experimented with CGVT, using raw image data without postprocessing on three challenging benchmarks—specifically, the Vaihingen dataset of the international society for photogrammetry and remote sensing (ISPRS) 2-DsSemantic labeling challenge [16], the Zeebrugge dataset of the 2015 Institute of Electrical and Electronics Engineers (IEEE) Geoscience and Remote Sensing Society (GRSS) data fusion contest (grss_dfc_2015) [17], and the LoveDA dataset [18] of the LoveDA semantic segmentation challenge. The proposed method achieved state-of-the-art performance on both Zeebrugge and LoveDA datasets and trailed on the Vaihingen dataset by 0.1%.

The remainder of this article is organized as follows. In Section II, we discuss related work. In Section III, we provide details on the proposed CGVT. We provide a description of the benchmark datasets and experimental settings in Section IV. In Section V, we present and analyze our experimental results. Concluding remarks are presented in Section VI.

II. Related Work

A. Remote Sensing Image Semantic Segmentation With Convolution

In recent years, with the rapid development of computing resources such as GPUs, deep learning technology has developed explosively. Deep learning technology has become the core method used for remote sensing image semantic segmentation. Because the deep convolutional network can automatically extract features without manual intervention, its weight sharing and translation invariance result in better trainability and generalization ability. Consequently, the deep convolution network is currently the most widely used deep

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

learning structure. Long et al. [19] proposed the fully convolutional network (FCN) and obtained the best results for semantic segmentation at that time. The FCN is a standard Encoder-Decoder structure; however, owing to the application of the FCN deconvolution layer, the boundaries of the obtained segmentation results are relatively coarse, which can especially damage the segmentation accuracy of VHR remote sensing images. Therefore, some scholars have refined the segmented coarse results based on the conditional random field (CRF) method by using context information to improve the segmentation accuracy. Sun and Wang [20] proposed a maximum hybrid strategy to combine deep semantic information and shallow detail information for VHR remote sensing image segmentation. Mou and Zhu [21] Proposed a new bidirectional network, called "recursive network" in full convolution network (RiFCN) to obtain a segmented image with more accurate boundary reasoning and semantic segmentation details. Ronneberger et al. [22] used the UNet decoder as a backbone, in combination with residual connections, dilated convolutions, pyramid scene parsing pooling, and multitasking inference to segment the VHR remote sensing image.

Context information is very important in semantic segmentation, especially long-distance context information. At first, researchers tried to improve the ability to obtain context information by increasing the receptive field. In this regard, Chen et al. [23] proposed the DeepLab network, which uses dilated convolution to increase the receptive field of common convolution kernels. The DeepLab family [24], [25] of networks has been pivotal in image semantic segmentation, forming a set of semantic segmentation methods with encoder-decoder as the basic paradigm. With the emergence of semantic segmentation methods with DeepLab as the basic encoder-decoder paradigm, many remote sensing scholars have done a lot of work on how to transfer these successful algorithms to remote sensing image segmentation. Yue et al. [26] proposed a new decoder called Tree-CNN block to fuse multiscale features and learn the best weight of the model for semantic segmentation of high-resolution images; Zheng et al. [27] propose a cross-modal decoder to collect multimodal image data for segmentation of high-resolution remote sensing images based on ResNet as the encoder; Sun et al. [28] also used ResNet as the decoder to extract high-level semantic features, and proposed multimodal fusion blocks (MFBs) to fully obtain the features of multimodal data for segmentation of high-resolution remote sensing images.

Although the context information can be increased by increasing the receptive field, it is, however, still insufficient for obtaining long-range context information. Therefore, increasingly more studies are applying CNNs based on the SA mechanism. For example, Li et al. [29] proposed spatial and channel attention modules that can refine features adaptively for VHR remote sensing images semantic segmentation. Ding et al. [30] proposed a patch attention module (PAM) and attention embedding module (AEM) to enhance the context information from the local and global levels for aerial images semantic segmentation.

The methods cited above are all based on CNN, and all use structures such as ResNet [31] as the backbone to obtain representation features. In contrast, in this article, we consider a different perspective—specifically, whether other methods can be used to replace the traditional CNN structure.

B. Transformer in Vision

The transformer [11] was proposed by the Google team in 2017 and applied in the NLP field to process sequential input data, such as natural language. Since then, the transformer has become the standard model for NLP. The SA mechanism is also a key application of the transformer model.

Inspired by the successful application of the transformer in NLP, an increasing number of researchers have recently tried to apply the transformer to the visual field. In the field of object detection, Carion et al. [32] proposed the DETR model, which applies the transformer to object detection, and surpassed the CNN method to become the best model at that time. In the field of image classification, Dosovitskiy et al. [12] proposed the ViT model, in which the transformer is applied to image classification. They were the first to use the pure transformer structure without any convolutional layers and obtained state-of-the-art results. Subsequently, Touvron et al. [33] proposed DeiT as an improvement on ViT that uses a novel knowledge distillation approach.

In the segmentation field, Wang et al. [34] proposed a new end-to-end video instance segmentation framework built upon transformers. Zheng et al. [35] proposed a pure transformer (i.e., without convolution and resolution reduction) to provide a powerful segmentation model, termed segmentation transformer (SETR). Recently, Liu et al. [13] proposed the Swin transformer, which is a hierarchical transformer whose representation is computed with shifted windows. The shifted windowing scheme results in greater efficiency by limiting SA computation to nonoverlapping local windows while also allowing for cross-window connection. The Swin transformer achieved state-of-the-art results on multiple CV tasks. Subsequently, many scholars were inspired by it to design a series of networks, such as [12], [13], [14], [15].

Recently, several applications based on the Swin transformer structure have appeared in the field of remote sensing. Gao et al. [36] fused the Swin transformer and CNN and proposed an adaptive fusion module (AFM) to fuse feature maps produced by the Transformer network and CNN. Xu et al. [37] used the Swin transformer as the backbone to explore the application of the transformer in remote sensing image semantic segmentation by combining different decoder heads.

Considering the studies cited above, it is clear that most transformer applications in remote sensing images are based on a successful transformer backbone for improvement. To better apply the transformer to the semantic segmentation of VHR remote sensing images, we designed a new end-to-end deep network called CGVT. In contrast to the methods cited above, CGVT is a brand-new backbone encoder framework based on the basic architecture of the transformer.

III. Proposed Method

To obtain long-range semantic context information, we restructure the deep network with the transformer framework to address the challenge of VHR aerial image semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

segmentation. The details of the proposed approach are presented in Sections III-A–III-E.

A. Overview of the Proposed Method

As shown in Fig. 1, the entire network is an encoder–decoder mechanism. In the encoder stage, the structure is the same as that of ordinary CNNs, such as ResNet. It is divided into four stages, and a set of feature maps is generated in each stage. There is a patch embedding module, a position embedding module, and a transformer encoder module in each stage. In the decoder stage, the feature maps generated by the encoder are fused, and then the segmentation result is generated.

As the input accepted by the transformer module is a 1-D input and our image is a 2-D input, we serialize the image. The input to the network is the image to be segmented, denoted $I \in \mathbb{R}^{C\times H\times W}$, C is the number of channels of the input images, and $(H \times W)$ is the original size of the input image. First, the input image I is divided into several nonoverlapping patches through the patch partition module. We set different patch sizes at different stages and record them as P_i, where i denotes the ith stage. Then, the size of the patches of different stages is $P_i \times P_i$. For the first stage, (N/P_i^2) patches can be obtained from the original image, where $i = 1$ and $N = HW$. This is the same as ViT processing.

Subsequently, we expand the obtained patches into a 1-D vector and then send the flattened patches into the patch embedding module. The patch embedding module is used for linear projection to map the flattened patches into D_i-dimensional embedding, then the shape of each embedded patch is $(N/P_i^2) \times D_i$, where $i = 1$. To obtain the spatial location information of image patches, it is necessary to encode the embedded patches and then feed them to the transformer encoder. The feature map output by the first stage is denoted as $F_1 \in \mathbb{R}^{D_1\times(H/P_1)\times(W/P_1)}$. In the same manner, the output of the previous stage is used as the input of the next stage to obtain the feature map of the different stages, F_2, F_3, F_4.

Through the above operations, the feature map of each stage is half the size of the feature map of the previous stage. The CNN adopts a similar method to reduce the size of the feature map, but it is realized through different convolution operation stripe sizes in the CNN.

We control the feature map size by changing the patch size of different stages. Specifically, for stage i, the output of the previous stage is denoted as $F_{i-1} \in D_{i-1} \times H_{i-1} \times W_{i-1}$, and we put F_{i-1} divided into P_i patches, then $F_i \in \mathbb{R}^{D_i\times(H_{i-1}/P_i)\times(W_{i-1}/P_i)}$, where $i = \{1, 2, 3, 4\}$, if $i = 1$, $(H_{i-1} \times W_{i-1})$ is equal to $(H \times W)$. Therefore, the feature map of each stage is P_i times smaller than that of the previous stage. In this way, the multiscale feature map similar to that obtained by ResNet is realized. Finally, F_i of each stage is sent to the decoder, and the final segmentation result is received.

B. Position Embedding

Following patch partitioning and embedding, the patches become a set of vectors without position information, which is permutation invariant. In other words, there is no spatial relationship between the features, which is extremely unfavorable for the segmentation of remote sensing images. Therefore, it is necessary to position-encode embedded patches. At present, the main position-encoding method involves configuring a set of learnable vectors to the same length as the input sequence [12], fixing them with sinusoidal functions [11], followed by absolute position encoding and relative position encoding [38].

The above methods, however, affect the results of semantic segmentation and destroy the translation variance of the network. We adopt the CPE proposed in CPVT [39] to replace ViT's position-encoding method improves performance. Therefore, in this study, we used CPE as our position-encoding method. CPE can dynamically generate position coding according to the local neighborhood of the input mark. This dynamic mechanism effectively avoids the shortcomings mentioned above.

CPE is generated by a position-encoding generator (PEG). The input to the generator comprises embedded patches through patch embedding, which is denoted as $E_i \in \mathbb{R}^{N\times D}$. Then, E_i is reshaped into $E_i' \in \mathbb{R}^{H'\times W'\times D}$ and CPE P_i is obtained by a 2-D convolution operation, $P_i = \mathrm{Conv}(E_i')$. It is worth noting that if the convolution kernel size is set to k, the $(k - 1/2)$ zero padding operation needs to be added, and $k = 3$ in this study. The zero padding operation is added because zero padding is the key to obtaining location information [40]. After the patch embedding and position embedding operations, the final image input sequence $S = E_i + P_i$.

C. Crisscross-Global Attention Transformer Encoder

Although the transformer has more powerful long-range context information acquisition ability than the convolution operation, it has the disadvantage being computation-intensive when applied to images, especially for intensive prediction tasks such as segmentation. At present, to reduce the impact of this problem, some studies [13], [41] reduce the amount of computation by applying SA to window; however, this method also affects the size of the receptive field, and the local attention affects the global feature expression of the image. Therefore, inspired by [14], [15], and [42], we designed the crisscross-global transformer encoder block (CG-TEB). CG-TEB can not only reduce the amount of calculation, but also consider the global feature expression.

As shown in Fig. 2, the crisscross-global transformer encoder is mainly composed of two different transformer encoder blocks: CC-TEB and GS-TEB.

1) Crisscross Transformer Encoder Block: To facilitate better comprehension of our CC-TEB module, we first present a standard transformer encoder block. A standard transformer encoder block is mainly composed of multihead SA (MSA), feed-forward network (FFN), multilayer perceptron (MLP), LayerNorm layer, and residual connections. As shown in Fig. 2(b), MSA is composed of multiple scaled dot-product attention. The scaled dot-product attention module is the specific way to realize SA, as shown in Fig. 2(c). It has three core concepts: query (Q), key (K), and value (V), where key

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

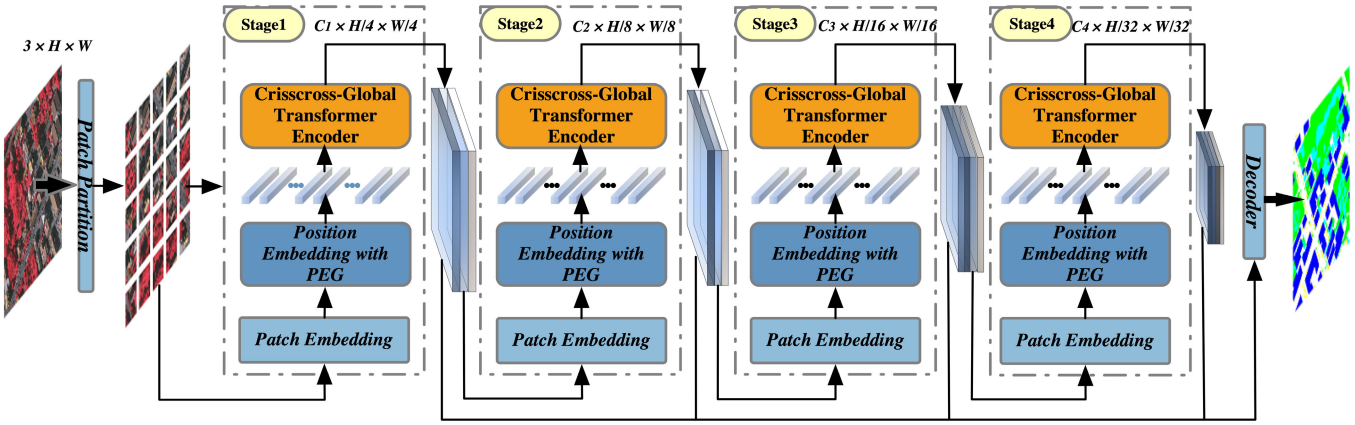


Fig. 1. Framework of the proposed CGVTs.

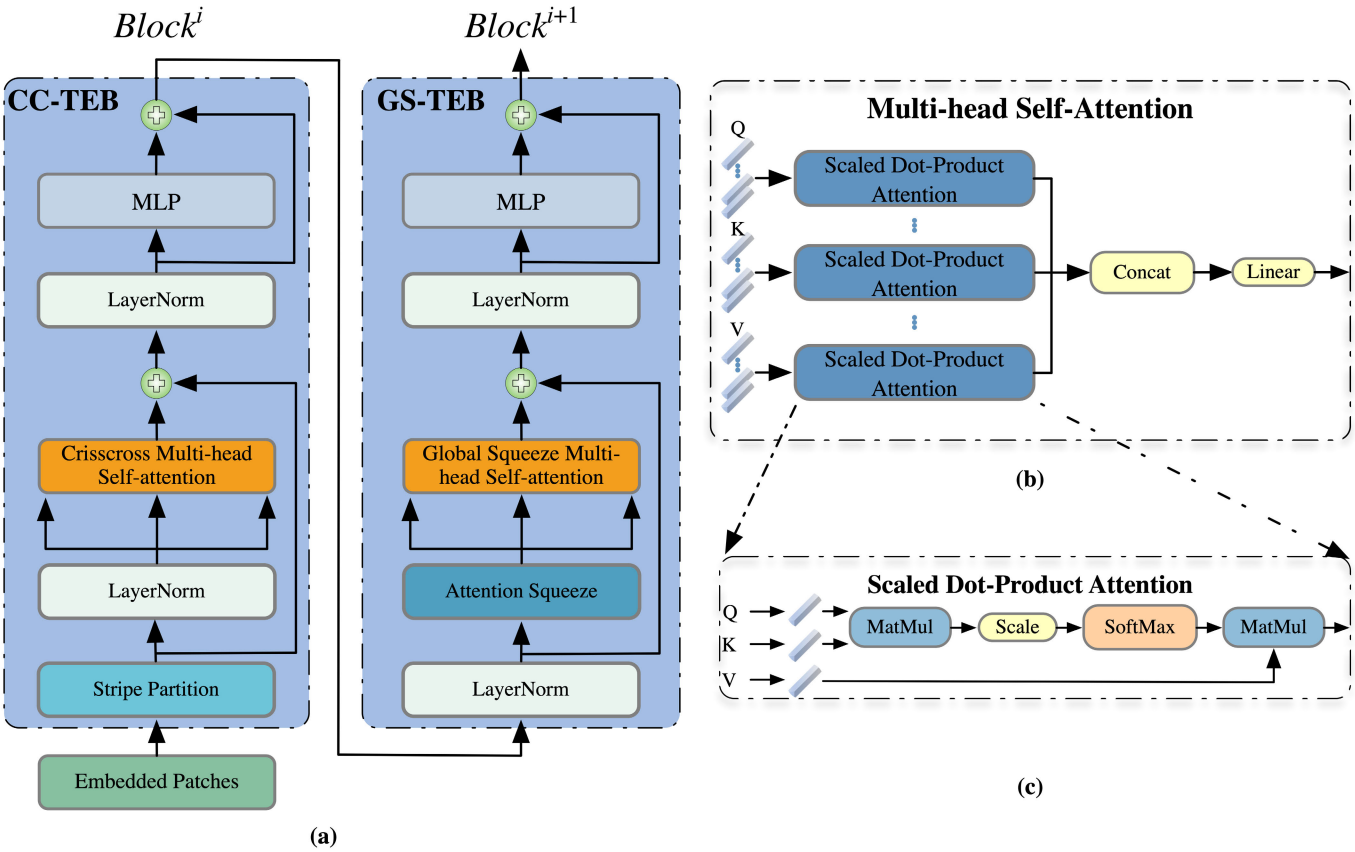


Fig. 2. (a) Two successive blocks of the crisscross-global transformer encoder: CC-TEB, crisscross transformer encoder block and GS-TEB, global squeeze transformer encoder block. (b) MSA structure. (c) Scaled dot-product attention structure.

and value are paired. Q, K, and V are obtained by embedded patches sequence $S \in \mathbb{R}^{N\times D}$ after linear embedding, where, N is the length of sequence S and D is the dimension of S. Then

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{1}$$

where W_Q, W_K, $W_V \in \mathbb{R}^{N\times D}$ are the learnable matrix parameters that pass through three linear projection layers. D is the dimension of Q, K, and V. SA is formulated as follows:

$$\mathrm{SA}(x) = \mathrm{softmax}\left(\frac{QK^T}{\mathrm{Scale}}\right)V \tag{2}$$

where $\mathrm{Scale} = \sqrt{D}$; the purpose of scale is to make the variance of the dot product of Q and K^T become one. If the scale operation is not carried out, the dot-product result will be very large. If that occurs, after softmax, the gradient will become very small and consequently disappear, which is not conducive to backpropagation.

MSA is an extension of SA, that is, the SA process is carried out k times. Finally, after concatenating the results of each SA, the final output is obtained through a linear projection, which is formulated as follows:

$$\mathrm{MSA}(x) = \mathrm{Concat}[\mathrm{SA}_1(x); \mathrm{SA}_2(x); \ldots; \mathrm{SA}_k(x)]W_O, \quad W_O \in \mathbb{R}^{k\cdot D_h \times D} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

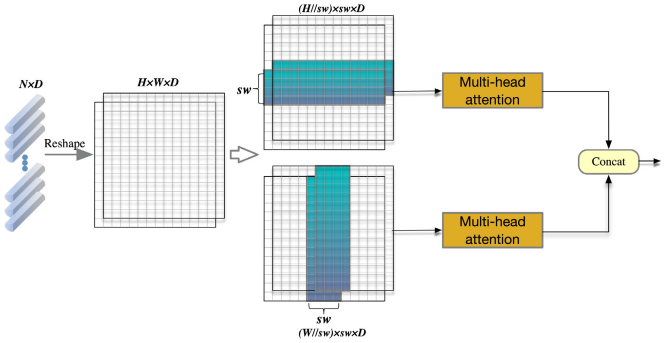


Fig. 3. Stripe partition operation.

where k denotes the number of SA operations, that is, the head number of MSA is k, W_O denotes the linear projections parameter, and Concat is the concatenate operation. Among them, D_h is typically set to (D/k).

MSA allows the attention layer to form multiple word spaces, which is similar to the function of multiple convolution kernels in the convolution operation. To focus the model on the information at different positions and extract different feature information, the MLP consists of two linear projection layers and uses a GELU.

In summary, a standard transformer encoder block Output can be formulated as follows:

$$S_\ell = \mathrm{MSA}(\mathrm{LN}(S_{\ell-1})) + S_{\ell-1}, \quad \ell = 1, 2, \ldots, L \tag{4}$$

$$S_\ell = \mathrm{MLP}(\mathrm{LN}(S_\ell)) + S_\ell, \quad \ell = 1, 2, \ldots, L \tag{5}$$

$$\mathrm{Output} = \mathrm{LN}(S_\ell) \tag{6}$$

where S_ℓ denotes the sum of the output after MSA and the input of the previous layers, LN denotes the LayerNorm operation, and ℓ denotes the ℓth layer.

Note that in Fig. 2(a), in the CC-TEB, there is a strip partition operation after embedded patches are fed into the block. Specifically, as shown in Fig. 3, given an embedded patches sequence $S \in \mathbb{R}^{N\times D}$, first, S is reshaped to $S \in \mathbb{R}^{H'\times W'\times D}$, then, S is divided into nonoverlapping vertical and horizontal stripes with a width of sw. If $H'\%\mathrm{sw} \neq 0$, $W'\%sw \neq 0$, then it is filled with padding, that is, we calculate SA in the stripe window. In this manner, vertical MSA (V-MSA) and horizontal MSA (H-MSA) are obtained. Note that V-MSA and H-MSA are calculated in parallel. Finally, the two kinds of attention are concatenated to obtain the final output of crisscross SA, denoted as $S_\ell^{\mathrm{CC_MSA}}$. Then, (4) and (5) can be revised to obtain as follows:

$$S_\ell^H = \mathrm{H-MSA}(\mathrm{LN}(S_{\ell-1})) + S_{\ell-1}, \quad \ell = 1, 2, \ldots, L \tag{7}$$

$$S_\ell^V = \mathrm{V-MSA}(\mathrm{LN}(S_{\ell-1})) + S_{\ell-1}, \quad \ell = 1, 2, \ldots, L \tag{8}$$

If the output of CC-TEB is recorded as $\mathrm{Output}_{\mathrm{CC-TEB}}$, then

$$S_\ell^{\mathrm{CC_MSA}} = \mathrm{Concat}(\mathrm{H-MSA}, \mathrm{V-MSA}) \tag{9}$$

$$S_\ell = \mathrm{MLP}\left(\mathrm{LN}\left(S_\ell^{\mathrm{CC_TEB}}\right)\right) + S_\ell^{\mathrm{CC_TEB}}, \quad \ell = 1, 2, \ldots, L \tag{10}$$

$$\mathrm{Output}_{\mathrm{CC-TEB}} = \mathrm{LN}(S_\ell). \tag{11}$$

2) Global Squeeze Transformer Encoder Block: Although crisscross MSA has less computation than the standard MSA, and increases the transformer's local representation ability, it also limits the transformer's original ability to obtain global long-range context information. In VHR remote sensing images, there is a strong context relationship between geographical entities. Therefore, the network needs to have global representation ability and a mechanism is needed to allow the information of different stripe windows to interact.

Based on the standard MSA, similar to [43], we propose global squeeze MSA (GS-MSA) to achieve the above goals. Similar to MSA, GS-MSA takes Q, K, and V as input vectors, but reduces the dimension of K and V. We call the dimension reduction operation, which reduces the computation amount, AS. As shown in Fig. 4, we use two different dimensionality reduction methods: linear adaptive average pooling, denoted as $\mathrm{AS}_{\mathrm{pooling}}$; and standard convolution layer reduction, denoted as $\mathrm{AS}_{\mathrm{conv}}$.

For the first method, as shown in Fig. 4(a), first $(K, V) \in \mathbb{R}^{N\times D}$ are reshaped into $(K, V) \in \mathbb{R}^{H\times W\times D}$. Then, the adaptive average pooling operation is performed; in this article, the pooling size $P = 7$. Next, the convolution operation is performed with kernel size $= 1$, which results in the dimension of (K, V) being $(K, V) \in \mathbb{R}^{P\times P\times D}$. Finally, after passing through the LayerNorm and GELU layers, it is fed into the SA operation. $\mathrm{AS}_{\mathrm{pooling}}$ can be formulated as follows:

$$\mathrm{AS}_{\mathrm{pooling}}(x) = \mathrm{LN}(\mathrm{Conv}1\times 1(\mathrm{Pooling}(\mathrm{Reshape}(K(x), V(x))))) \tag{12}$$

where x denotes the embedded patches vectors, Reshape is the input reshaping operation $x \in \mathbb{R}^{N\times D}$, Pooling is the adaptive average pooling operation, conv 1×1 denotes the convolutional operation with kernel size 1×1, and LN is the LayerNorm operation.

For the second method, as shown in Fig. 4(b), as in the first method, $(K, V) \in \mathbb{R}^{N\times D}$ are first reshaped into $(K, V) \in \mathbb{R}^{H\times W\times D}$. Then, the dimension is squeezed directly through a convolution layer. Note that similar to the dilation rate of dilated convolution, here we control the compression degree by setting different squeeze rates, and the stripe setting is the same as the squeeze rate, so that the compressed space does not overlap. $\mathrm{AS}_{\mathrm{conv}}$ is formulated as follows:

$$\mathrm{AS}_{\mathrm{conv}}(x) = \mathrm{LN}(\mathrm{Conv}(\mathrm{Reshape}(K(x), V(x)))) \tag{13}$$

where x denotes embedded patches vectors, Reshape is the reshaping input operation $x \in \mathbb{R}^{N\times D}$, Conv denotes the convolutional operation with kernel size $=$ squeeze rate, and LN is the LayerNorm operation.

In summary, (2)–(4) can be revised to obtain as follows:

$$\mathrm{AS-SA}(x) = \mathrm{softmax}\left(\frac{\mathrm{Q(AS(K))}^{\mathrm{T}}}{\mathrm{Scale}}\right)\mathrm{AS}(V) \tag{14}$$

$$\mathrm{GS-MSA}(x) = \mathrm{Concat}[\mathrm{AS-SA}_1(x); \mathrm{AS-SA}_2(x); \ldots; \mathrm{AS-SA}_k(x)]W_O, \quad W_O \in \mathbb{R}^{k\cdot D_h\times D} \tag{15}$$

$$S_\ell = \mathrm{GS-MSA}(\mathrm{LN}(S_{\ell-1})) + S_{\ell-1}, \quad \ell = 1, 2, \ldots, L \tag{16}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

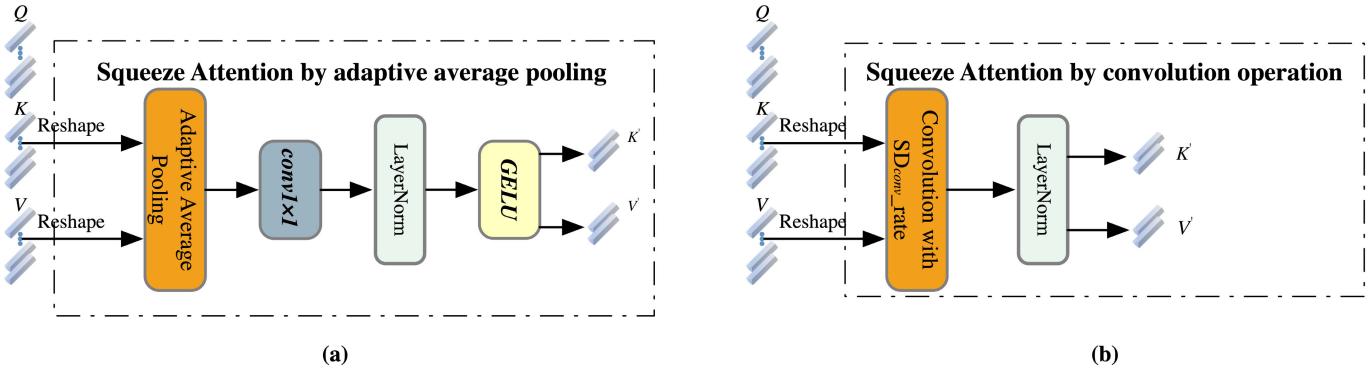


Fig. 4. (a) Structure of squeeze attention based on adaptive average pooling ($AS_{pooling}$). (b) Structure of squeeze attention based on convolution operation.

where $AS-SA$ denotes the SA after AS, x denotes the embedded patches vectors, AS denotes the AS operation, and $GS-MSA$ denotes global squeeze MSA. If the output of the GS-TEB is recorded as Output_{GS-TEB}, then

$$S_\ell = GS-MSA(LN(S_{\ell-1})) + S_{\ell-1}, \quad \ell = 1, 2, \ldots, L \tag{17}$$

$$S_\ell = MLP(LN(S_\ell)) + S_\ell, \quad \ell = 1, 2, \ldots, L \tag{18}$$

$$\text{Output}_{GS-TEB} = LN(S_\ell). \tag{19}$$

3) Crisscross-Global Attention Transformer Encoder: As illustrated in Fig. 2(a), the crisscross-global transformer encoder is composed of two successive blocks: CC-TEB and GS-TEB. The workflow is as follows: input embedded patches into CC-TEB and obtain Output_{CC-TEB}. Then, use Output_{CC-TEB} as input to GS-TEB to get Output_{GS-TEB}.

D. Decoder

To evaluate the feature expression ability of crisscross-global attention transformer encoder, we used two different decoder structures: feature pyramid network (FPN) [44] and UperNet [45]. These two decoders are relatively simple structures, do not contain the divided dilated convolution operation, and can fully use the features of different stages.

1) FPN Decoder: FPN can fuse multiple spatial resolution features. If the output of each stage of CGVT Encoder is recorded as $F_i \in \mathbb{R}^{D_i \times (H_{i-1}/P_i) \times (W_{i-1}/P_i)}$, then the input of FPN are $\{F_1, F_2, F_3, F_4\}$. The FPN decoder is divided into two parts, as shown in Fig. 5(a). In the first part, feature maps with different depths are unified into the same depth C via a bottleneck layer (1×1 convolution-BN-ReLU) through the path from top to bottom (the depth C is set to 256 in the standard FPN), and the output is $F'_i \in \mathbb{R}^{256 \times (H_{i-1}/P_i) \times (W_{i-1}/P_i)}$. In the second part, with $F'_{i=4}$ as input, starting from the deepest FPN feature level (F'_4), $2\times$ upsampling is performed to upsample it to the size of F'_{i-1}, then element-wise sum with F'_{i-1}; each upsampling operation includes 3×3 convolution, group norm [46], ReLU, and $2\times$ bilinear upsampling. Then, the same sequence of operations is performed on F'_3, F'_2, and F'_1 perform. Since the size of each feature map is increased by twice, the size of final output is $1/P_1$ scale feature maps. Finally, 1×1 convolution, $P_1 \times$ bilinear upsampling, and softmax are applied to obtain the segmentation results of $H \times W$ size.

2) UperNet Decoder: The UperNet decoder is based on FPN, as shown in Fig. 5(b). Thus, the UperNet decoder is also divided into two parts. The first part is similar to FPN. If the output is $F'_i \in \mathbb{R}^{C \times (H_{i-1}/P_i) \times (W_{i-1}/P_i)}$. It is worth noting that F_4 is fed into the pyramid pooling module (PPM) from PSPNet [68] to obtain F'_4. In the second part, similar to FPN, starting from the deepest feature level F'_4, we resize the size of F'_4 to the size of F'_1, the same operations for $\{F'_3, F'_2\}$ and finally concatenate these feature maps. Finally, 1×1 convolution, $P_1 \times$ bilinear upsampling, and softmax are applied to obtain the segmentation results of $H \times W$ size.

E. Architecture Variants

In this section, we list several important hyperparameters of our network. As in ViT, we can easily scale up our method based on different hyperparameters, namely, CGVT-Small, CGVT-Medium, and CGVT-Large, abbreviated as CGVT-S, CGVT-M, and CGVT-L, respectively. As in the network structure design of ResNet, CGVT has four stages in total, and with increasing network depth, the dimension of the hidden layer increases, and the output resolution decreases gradually. The detailed architectural specifications are presented in Table I, where "$H \times W$" denotes the size of the input image and "stripe. sz. =7" denotes an MSA module with a stripe size of seven.

P_i: Patch size of the stage i.
D_i: Channel number of the output of stage i.
N_i: Head number of CC-MSA and GS-MSA in stage i.

IV. Experimental Evaluation

A. Datasets

We experimented on three widely used open VHR aerial image datasets: Vaihingen [47] (in the ISPRS 2-D semantic labeling benchmark dataset), Zeebrugge [48], and LoveDA [18]. These three image datasets consist of VHR true orthophoto (TOP) tiles. Specifically, the resolution of the images in the Vaihingen and Zeebrugge datasets is at the centimeter level and that of those in the LoveDA dataset is at the submeter level.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

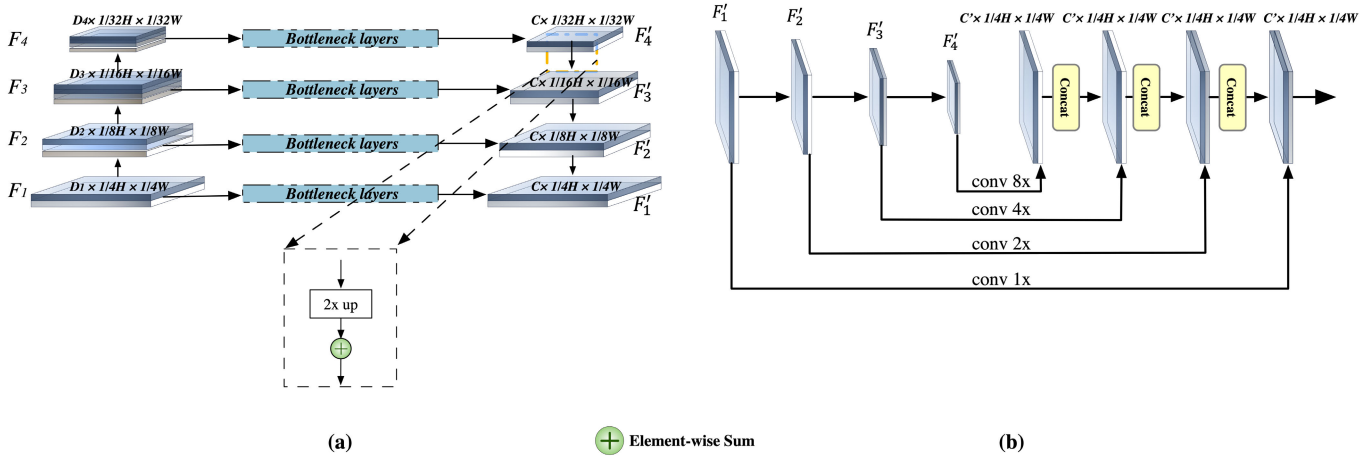


Fig. 5. (a) Structure of feature pyramid network (FPN) decoder. (b) Structure of UperNet decoder.

TABLE I
DETAILED ARCHITECTURAL SPECIFICATIONS FOR CGVT

	Downsp. Rate (output size)	Layer Name	CGVT-S		CGVT-M		CGVT-L	
Stage 1	4x H/4×W/4	Patch Embedding	P_1=4; D_1=64		P_1=4; D_1=96		P_1=4; D_1=128	
		Transformer	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =1}\\ D_1=64;\ N_1=2\end{bmatrix}\times1$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =1}\\ D_1=96;\ N_1=3\end{bmatrix}\times1$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =1}\\ D_1=128;\ N_1=4\end{bmatrix}\times1$
		Encoder	GS-TEB	$[D_1=64;\ N_1=2]\times1$	GS-TEB	$[D_1=96;\ N_1=3]\times1$	GS-TEB	$[D_1=128;\ N_1=4]\times1$
Stage 2	8x H/8×W/8	Patch Embedding	P_2=2; D_2=128		P_2=2; D_2=192		P_2=2; D_2=256	
		Transformer	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =2}\\ D_2=128;\ N_2=4\end{bmatrix}\times1$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =2}\\ D_2=192;\ N_2=6\end{bmatrix}\times1$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =2}\\ D_2=256;\ N_2=8\end{bmatrix}\times1$
		Encoder	GS-TEB	$[D_2=128;\ N_2=4]\times1$	GS-TEB	$[D_2=64;\ N_2=6]\times1$	GS-TEB	$[D_2=256;\ N_2=8]\times1$
Stage 3	16x H/16×W/16	Patch Embedding	P_3=2; D_3=256		P_3=2; D_3=384		P_3=2; D_3=512	
		Transformer	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_3=256;\ N_3=8\end{bmatrix}\times5$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_3=384;\ N_3=12\end{bmatrix}\times9$	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_3=512;\ N_3=16\end{bmatrix}\times9$
		Encoder	GS-TEB	$[D_3=256;\ N_3=8]\times5$	GS-TEB	$[D_3=384;\ N_3=12]\times9$	GS-TEB	$[D_3=512;\ N_3=16]\times9$
Stage 4	32x H/32×W/32	Patch Embedding	P_4=2; D_4=512		P_4=2; D_4=768		P_4=2; D_4=1024	
		Transformer	CC-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_4=512;\ N_4=16\end{bmatrix}\times2$	GS-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_4=768;\ N_4=24\end{bmatrix}\times1$	GS-TEB	$\begin{bmatrix}\text{stripe. sz. =7}\\ D_4=1024;\ N_4=32\end{bmatrix}\times1$
		Encoder	GS-TEB	$[D_4=512;\ N_4=16]\times2$	GS-TEB	$[D_4=768;\ N_4=24]\times1$	GS-TEB	$[D_4=1024;\ N_4=32]\times1$

1) Vaihingen Dataset: The images comprising the dataset were captured over Vaihingen in Germany by the German Association of Photogrammetry and Remote Sensing (DGPF). The dataset is an urban scenario, of which the primary ground object classes are individual multistory buildings, roads, and cars. The dataset consists of 33 orthophoto patches with an average size of 1388 × 3313 pixels. Sixteen images in the dataset have corresponding labels and can be used for training, and the remaining 17 images are undisclosed by organizers for testing. The patches are 8 bit TIFF files with three bands; the three RGB bands of the TIFF files correspond to the near infrared, red, and green bands delivered by the digital aerial cameras, with a ground sample distance (GSD) of 9 cm. The ground object classes and corresponding ground truth colors in the dataset are as follows: impervious surfaces (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red).

Because of the very high resolution of the dataset, the ground objects in the images are very detailed, which results in a huge challenge for accurate segmentation. The main difficulties are as follows. First, the geographic entities have highly detailed shapes, geographical scenes have very

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

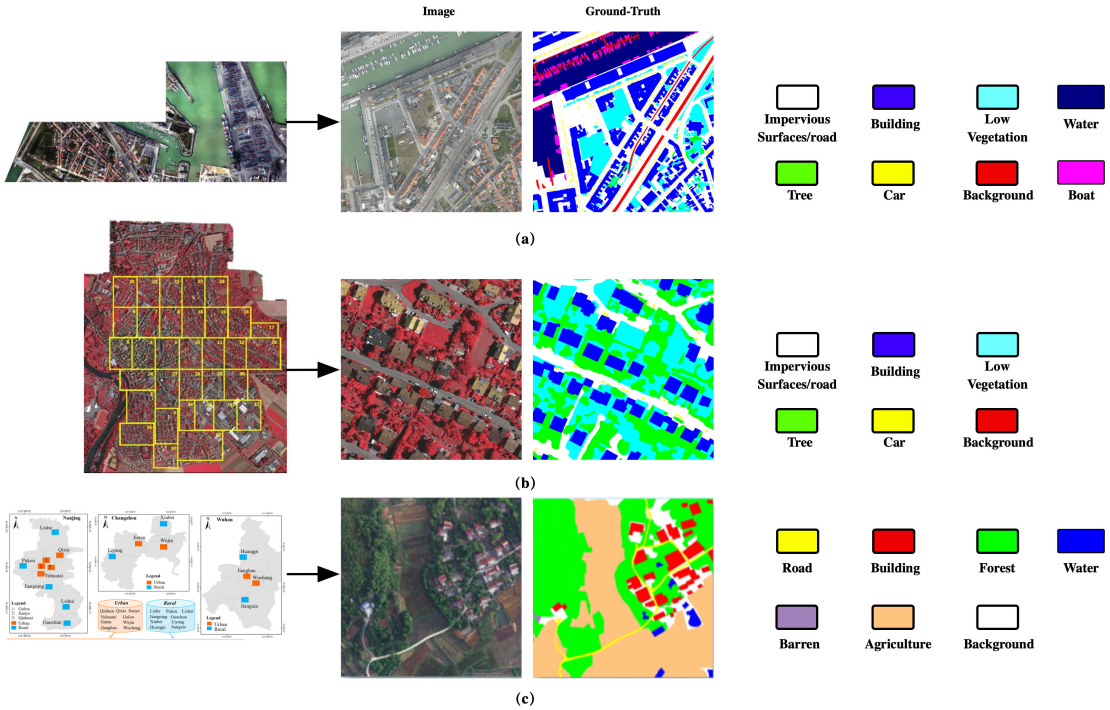


Fig. 6. Examples of datasets for original images and their reference images in our experiments. (a) Vaihingen dataset. (b) Zeebrugge dataset. (c) LoveDA dataset.

complicated content, and geographical elements have multiscale morphological characteristics. Second, the ground objects are not evenly distributed in the dataset: the pixels of "buildings" and "impervious surfaces" are 54.09% of the overall label pixels, whereas the pixels of "cars" are 1.21%. The pixel distribution for the ground truths can be seen in Table II. In this experiment, we used 11 images with labels as the training set, and the remaining five images as validation set (IDs 11, 15, 28, 30, 34) to acquire basic hyperparameters. Subsequently, we used all 16 images with ground truths as training data and the remaining 17 images in the dataset as the test set to evaluate the feature expression and generalization abilities of the model. Examples of origin images and respective ground truths of the Vaihingen dataset are shown in Fig. 6(a).

2) Zeebrugge Dataset (grss_dfc_2015): This dataset is from the 2015 IEEE GRSS Data Fusion Contest, and was acquired in the visible wavelength range. The imaging data were obtained via an airborne platform flying at an altitude of 300 m over the urban and harbor areas of Zeebrugge, Belgium. The images are of the urban and harbor areas of Zeebrugge. The whole dataset consists of seven orthophoto images with a size of 10 000 × 10 000, five of which have ground truth (ID Tiles 1, 2, 3, 5, 7), and the remaining two images (ID Tile 4 and Tile 6) are undisclosed by organizers for testing. These tiles are 8 bit TIFF files with R, G, B bands, and a GSD of 5 cm. The ground object classes and the corresponding ground truth colors in the dataset are as follows: impervious surface (white), buildings (blue), low vegetation (cyan), trees (green), cars (yellow), clutter (red), boats (pink), and water (dark blue).

Compared with the Vaihingen dataset, the Zeebrugge dataset has richer details because of the higher spatial resolution of the images, and the uneven distribution of ground object classes is more severe. The pixels of "cars" and "boats" are only 0.5% and 0.7% of the overall pixels. These characteristics of the dataset make segmentation more challenging.

In this experiment, we used four of the five images with corresponding labels as the training dataset, and the remaining one image as the validation set to determine appropriate hyperparameters. Subsequently, all five images with ground truths were used as training data. The remaining two images in the dataset were used as the test set, and were uploaded to the evaluation website for evaluation. Examples of origin images and respective ground truths of the Vaihingen dataset are shown in Fig. 6(b).

3) LoveDA Dataset: This dataset is the latest open dataset for remote sensing image semantic segmentation, and is used in challenging competitions. The dataset images were acquired from Google Earth and depict rural and urban areas from three different cities in China. The spatial resolution of the dataset is 0.3 m and the total area coverage is 536.15 km^2. The entire image dataset consists of 5987 geographically rectified and preprocessed nonoverlapped images with a resolution of 1024 × 1024 pixels. Among the images in the dataset, 4191 are labeled, 2522 of them served as training data, and 1699 served as the validation dataset. The remaining 1796 images served as testing data and the results were uploaded to the website for challenging competitions. The ground object classes and the corresponding ground truth colors in the dataset are as follows: buildings (red), road (yellow), water (blue), barren (purple), forest (green), agricultural (orange), background (white).

The LoveDA dataset has a low resolution compared with the two aforementioned datasets; specifically, at the submeter

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NUMBER OF PIXELS AND THE PROPORTION PER CLASS FOR THE VAIHINGEN AND ZEEBRUGGE DATASETS

	Vaihingen [3]		Zeebrugge [17]	
Class	**#Pixels**	**Proportion (%)**	**#Pixels** (MPix)	**Proportion (%)**
Impervious Surfaces/Road	21815349	27.94	235.5	33.6
Building	20417332	26.15	57.6	8.2
Low Vegetation	16272917	20.84	75.5	10.8
Tree	18110438	23.19	13.9	2.0
Car	945687	1.21	3.2	0.5
Clutter/Background	526083	0.67	54.8	7.8
Boat	—	—	4.7	0.7
Water	—	—	201.0	28.7

TABLE III
NUMBER OF PIXELS AND THE PROPORTION PER CLASS FOR THE LOVEDA DATASET

	LoveDA [18]	
Class	**#Pixels**	**Proportion (%)**
Road	332923846	54.73
Building	616243956	10.13
Forest	716311869	11.78
Barren	261880400	4.31
Agriculture	1544771605	25.39
Background	1994496285	32.79

level. There are more pixel samples and labeled targets in each class, and such complex samples, however, result in higher intraclass variance, and therefore, more severe misclassifications. In addition, the dataset contains urban and rural areas. There are differences in the ground object distributions between urban and rural areas; for example, urban areas have more man-made objects such as roads and buildings, whereas rural areas have more agricultural land. Such ground object differences are a major challenge to the generalization ability of the model. Moreover, the dataset has uneven ground object distribution, as shown in detail in Table III, but it does not contains fine-grained objects, such as cars and boats. Examples of origin images and respective ground truths of the Vaihingen dataset are shown in Fig. 6(c).

B. Experiment Settings

1) Experimental Environment: All the experiments were conducted on a cluster computer system with four Tesla V100-SXM2 (16 GB RAM) GPUs. The framework was implemented using the PyTorch open source toolbox [49].

2) Data Preprocessing: First, we conducted data augmentation on the original dataset. This comprised two stages: 1) cropping and rotating every image in the original dataset. We used a slide window with size of 512 × 512 to crop the original images from top left to bottom right with some overlap. The overlap rate was 50%. Then, the small image tiles were flipped horizontally and vertically, and rotated by $\theta^\circ = 90$ and 2) in the training process, we took the image data from the first stage as input data and performed scaling (up or down) of the images, as well as random cropping. Meanwhile, we distorted the current images using several photometric methods, such as random brightness and random contrast. To increase the convergence speed of the network, we conducted mean normalization and standardization on the input data. The detailed method was as follows:

$$x^{\text{normaliztion}} = \frac{x - \text{mean}(x)}{\max(x) - \min(x)} \tag{20}$$

where $\text{mean}(x)$, $\max(x)$, and $\min(x)$ are the mean, maximum, and minimum values of the data, respectively,

$$x^{\text{standardizaiton}} = \frac{x - \mu}{\sigma} \tag{21}$$

where μ, σ are the average and standard deviation values of the sample data, respectively.

3) Implementation Details: When applying deep learning methods to segmentation tasks, to increase the convergence speed, a large dataset is typically required to train a set of parameters, for example, ImageNet; then, the parameters are applied to the segmentation tasks. Those parameters are called pretrained parameters. When using CNNs, most methods use ResNet as the backbone for encoding, and ResNet has public pretrained parameters. Our network is based on a new network structure, the "Transformer," and there is no public pretrained parameters for it. Consequently, we trained our network using the ImageNet dataset to acquire pretrained parameters and then used them in the segmentation tasks.

The ImageNet dataset is a huge computer visualization dataset. The most commonly used datasets are ImageNet-1K and ImageNet-22K. The former contains 1.28 million training images and 50 000 validation images in 1000 classes. The latter is larger, containing 14.2 million images in 220 000 classes. Because of hardware limitations, we used the ImageNet-1K dataset in our study.

With the same experimental setup as in [13] and [39], we trained models with mini-batch stochastic gradient descent and the AdamW [50] optimizer with an initial learning rate of 5×10^{-4} for 200 epochs, and weight decay set to 0.05. The batch size was set to 128 images for every GPU. For the learning rate scheduler, we employed cosine annealing decay and 20 epochs of linear warm-up.

As with normal CNNs, CGVT was trained end-to-end. With enough hardware resources, CGVT can take images of any size as input for training and inference. We used the patches of 512 × 512 pixels as input size. In the training and validation phases, we conducted center cropping on images, with the size set to 480 × 480 pixels.

4) Evaluation Metrics: For fair comparison, we used different evaluation metrics for the three different datasets. For the Vaihingen dataset, we used the ISPRS benchmark test,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

with the overall accuracy (OA) and F1 score of each class as our evaluation metrics. F1 score can be computed in terms of completeness (recall) and correctness (precision) (please refer to the equations for more detail). In our evaluation, the areas that were three pixels around the boundaries were neglected. Those eroded areas were then ignored during evaluation to eliminate the influence of uncertain boundaries.

For the Zeebrugge dataset, we used the evaluation metrics used in the 2015 IEEE GRSS Data Fusion Contest: OA, average accuracy (AA), kappa index ($\mathcal{K}$), and F1 score.

Recall, precision, F1, Kappa index, AA, and OA are, respectively, defined as follows:

$$\text{Precision} = \frac{\text{tp}}{\text{tp}+\text{fp}} \quad \text{Recall} = \frac{\text{tp}}{\text{tp}+\text{fn}}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

$$\text{OA} = \frac{\sum_{i=1}^{M} N_{ii}}{N} \quad \text{AA} = \frac{\text{Sum(Recall)}}{M} \tag{23}$$

where tp, fp, and fn are the number of true positives, false positives, and false negatives, respectively. N is the total number of pixels and M is the number of classes. $\sum_{i=1}^{M} N_{ii}$ is the sum of the number of pixels on the diagonal of the confusion matrix.

$$\mathcal{K} = \frac{\mathcal{P}_o - \mathcal{P}_e}{1 - \mathcal{P}_e} \mathcal{P}_e = \frac{1}{N^2} \sum_{k} n_{k1} n_{k2} \tag{24}$$

where $\mathcal{P}_o$ is the proportion of observation accuracy or consistency unit, equivalent to OA; and $\mathcal{P}_e$ is the proportion of expectant accidental consistent units. For k categories, N is the number of observations needed to be categorized and n_{ki} is the number of times that i predicts category k.

For the LoveDA dataset, we used the evaluation standards of the LoveDA Semantic Segmentation Challenge. Thus, we adopted mean intersection over union (mIoU), which is the average of IoU, as our evaluation metric. The equations are as follows [51]:

$$\text{IoU} = \frac{p_{ii}}{\sum_{j=0}^{M} p_{ij} + \sum_{j=0}^{M} p_{ji} - p_{ii}} \tag{25}$$

$$\text{mIoU} = \frac{1}{M+1} \times \sum_{M} \text{IoU}_M \tag{26}$$

where M is the number of classes and p_{ij} is the number of pixels of class i inferred to belong to class j. In other words, p_{ii} represents the number of true positives and p_{ij} and p_{ji} denote false positives and false negatives, respectively.

V. Results and Discussion

In this section, we demonstrate the effectiveness and generalization of our proposed CGVT. First, we compare the proposed method with the state-of-the-art methods in the relevant published or preprint papers in which the three datasets are also used. Next, we analyze the effect of different model scales and decoders and the effect of pretraining with ImageNet. Then, we present an ablation study conducted to evaluate the effect of the submodules.

A. Comparison With State-of-the-Art Methods

1) Results on Vaihingen Dataset: We compared the achieved results with those from the different methods of the Vaihingen dataset on the ISPRS semantic labeling contest (2-D) website or published papers. The basic architecture configurations of the methods are summarized in Table IV.

Quantitative and qualitative comparisons of the results obtained on the Vaihingen dataset are shown in Table V and Fig. 7, respectively. For the quantitative perspective, many methods were compared, as shown in Table V. The results show that our OA was 92.0% with only the original image data for training. This is the joint second result, shared with G2GNet, which imports auxiliary data, such as NDVI and NDSM. In terms of F1 measure, our method achieved satisfactory results in each class. In particular, we achieved high performance on fine-grained objects and man-made objects.

It is worth noting that we divided these methods into structures based on CNNs and structures based on transformer in Table IV, and we are aware that the methods based on CNN are still the mainstream deep learning partitioning methods. In contrast, the transformer-based methods are mostly based on the Swin transformer structure. The segmentation results of STransFuse are not particularly satisfactory. Although the DCFAM method performs well in terms of OA, it is also slightly inadequate on fine-grained objects such as car classes. Our method achieved more satisfactory results in terms of both the overall OA and the accuracy of each class. In particular, the proposed method outperforms all CNN-based and transformer-based methods in terms of OA by a large margin.

Fig. 7 gives a qualitative comparison of the results. We selected several sets of segmentation result maps on the evaluation website for comparison with our results. Generally, it can be observed that the segmentation maps obtained by our method have smoother boundaries and the phenomenon of spectral heterogeneity of the same geographic entities increases and the spectral difference between different geographic entities decreases in VHR images are partially alleviated. Specifically, the Vaihingen dataset has numerous artificial objects, such as roofs and roads, and some white cars, which are very prone to misclassification.

For row 1 in Fig. 7, the shadow between two buildings is difficult to distinguish. The actual segmentation results show that other methods would mistakenly classify the shadow part as road, whereas our method successfully segments the object to a certain degree. In row 2–5, because the color of the building roof is similar to that of the roads and cars, most of the methods did not achieve good segmentation results; however, our method almost achieved a more desirable segmentation result.

2) Results on Zeebrugge Dataset: For the Zeebrugge dataset we also evaluated the results from both quantitative and qualitative aspects. Quantitative comparison results for the Zeebrugge dataset are shown in Table VI, including the results submitted to the GRSS data and algorithm standard evaluation website and reported in published or preprint papers. Because the metric for ranking on the GRSS Data and algorithms criteria assessment website are evaluated with AA, we use

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
INFRASTRUCTURE CONFIGURATION OF THE EXPERIMENTAL COMPARISON METHODS ON THE VAIHINGEN DATASET

Category	Method	Structure
CNN-based	UZ_1 [16]	Standard CNN + deconvolutions + NDSM
	SSAtNet [52]	ResNet101 + pyramid attention pooling module
	HCANET [53]	ResNet101 + compact atrous spatial pyramid pooling module
	ONE_7 [54]	SegNet + NDSM
	MFNet [28]	ResNet50 + NDSM + two branches
	TreeUNet [26]	ResNet101 + tree-CNN block
	SBANet [55]	ResNet101 + boundary attention module
	MAResU-Net [56]	ResNet34 + linear attention mechanism
	BKHN11 [16]	ResNet101
	CASIA2 [4]	ResNet101 + self-cascaded architecture
	CCANet [57]	ResNet101 + CCA module
	NLPR3 [16]	ResNet101
	HUSTW5 [58]	ResNet101 + structural stereotype
	G2GNet [27]	ResNet101 + NDSM + NDVI
	SLGENet-3 [59]	CNN + NAS
Transformer-based	DCFAM [60]	Swin transformer + densely connected feature aggregation module
	STransFuse [36]	Swin transformer + feature fusion
	CGVT (ours)	Crisscross-Global Attention Transformer (CGVT)

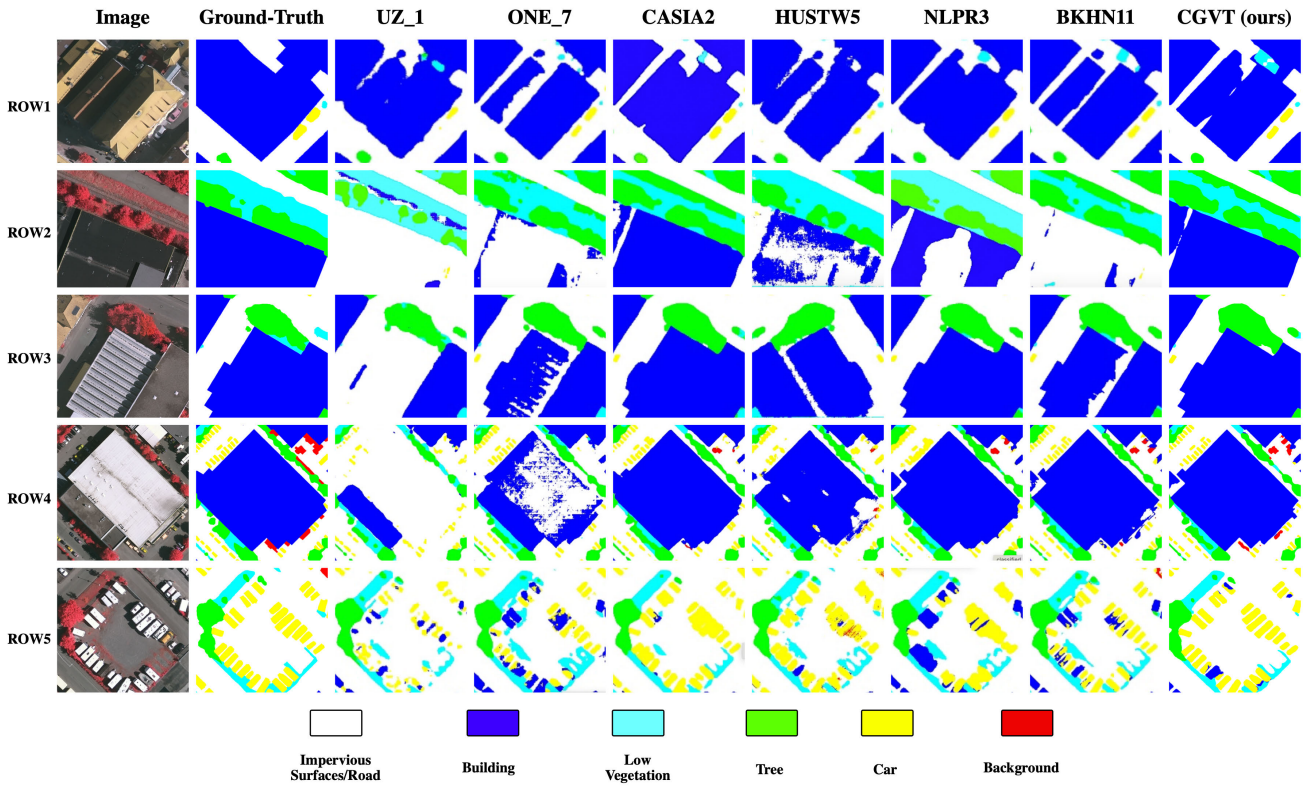


Fig. 7. Qualitative comparison between the proposed CGVT and official results on the Vaihingen dataset.

AA as the ranking criterion in our table for the Zeebrugge dataset.

From the quantitative perspective, our method is ranked first on the evaluation website. Compared with the baseline [17],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED CGVT AND OTHER OFFICIAL OR PUBLISHED RESULTS FOR THE VAIHINGEN DATASET. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%) ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/Road	Building	Low Vegetation	Tree	Car	OA
STransFuse [36]	88.3	91.5	79.0	85.5	77.1	86.1
UZ_1 [16]	89.2	92.5	81.6	86.9	57.3	87.3
SSAtNet [52]	90.1	93.7	80.3	86.5	81.2	88.0
HCANET [53]	92.2	95.6	80.7	88.9	87.4	89.7
ONE_7 [54]	91.0	94.5	84.4	89.9	77.8	89.8
MFNet [28]	92.2	96.6	81.7	89.1	**89.7**	90.2
TreeUNet [26]	92.5	94.9	83.6	89.6	85.9	90.4
SBANet [55]	**94.4**	93.0	83.4	89.6	91.4	90.6
MAResU-Net [56]	92.9	95.2	84.9	89.9	88.3	90.7
BKHN11 [16]	92.9	96.0	84.6	89.9	88.6	91.0
CASIA2 [4]	93.2	96.0	84.7	89.9	86.7	91.1
CCANet [57]	93.3	94.3	82.0	88.6	86.6	91.1
NLPR3 [16]	93.0	95.6	85.6	90.3	84.5	91.2
HUSTW5 [58]	93.3	96.1	**86.4**	**90.8**	74.6	91.6
DCFAM [60]	93.6	96.2	85.7	90.4	87.6	91.6
G2GNet [27]	93.7	**96.5**	86.1	90.8	88.2	92.0
SLGENet-3 [59]	94.0	95.7	84.8	**91.5**	89.4	**92.1**
CGVT-Large (ours)	94.1	95.9	84.2	90.0	88.5	92.0

TABLE VI

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED CGVT AND OTHER OFFICIAL RESULTS FOR THE ZEEBRUGGE DATASET. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%), AA (%), AND KAPPA ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/ Road	Water	Clutter	Low Vegetation	Building	Tree	Boat	Car	OA	AA	Kappa
VGG/SVM [17]	67.66	96.5	45.6	68.38	72.7	78.77	56.1	33.9	76.6	64.95	0.7
AlexNet [17]	79.1	98.2	63.4	78	75.6	79.5	44.8	50.8	83.32	72.52	0.78
DeepLab v3 [35]	84.92	98.62	49.76	80.00	80.03	79.37	48.55	80.61	87.32	75.23	0.77
RotEqNet [61]	77.41	98.69	48.99	78.93	79.07	71.3	44.5	52.54	82.6	75.3	0.77
DenseASPP [62]	79.42	96.98	61.21	77.86	78.82	75.63	64.51	81.11	84.35	76.94	0.82
SegNet (+SDT) [63]	84.04	98.94	**69.25**	83.88	81.71	80.04	50.83	80.27	87.31	78.62	—
DNN_HCRF [64]	—	—	—	—	—	—	—	—	87.85	—	0.84
RIT [58]	84.8	98.97	62.83	84.24	83.93	80.17	55.77	83.13	87.91	79.23	0.84
MFNet [28]	86.95	99.03	61.72	81.77	84.55	77.52	69.29	88.65	88.87	—	—
CCANet [57]	87.11	98.75	61.45	**85.68**	82.95	80.36	70.25	88.05	88.83	81.83	0.85
CGVT-Large (ours)	**89.46**	**99.06**	66.87	82.65	**87.56**	**83.39**	**76.39**	**91.34**	**89.86**	**84.59**	**0.87**

AA increases from 72.52% to 84.59%, OA from 83.32% to 89.86%, and Kappa from 0.78 to 0.87. Compared to the latest literature results [28], [57], our AA represents an improvement of 2.76%, OA by 0.99%, and Kappa by 0.02. In terms of F1 measures for each class and the accuracy for each class, our method achieved state-of-the-art performance. In particular, for the car and boat classes, which are very challenging owing to severe class imbalance, our accuracy results are much better than those of the latest methods. Specifically, the F1 scores of Boat rises from 70.25% to 76.39% in comparison with [57], and the F1 scores of car from 88.65% to 91.34% in comparison with [28]. Thus, from the quantitative perspective, our results are outstanding.

Simultaneously, from a qualitative perspective, because the GRSS Data and Algorithm Standard Evaluation website shows only the quantitative results of the competitors, we were unable to qualitatively compare our results with those of other methods. Therefore, we only compared the results employing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

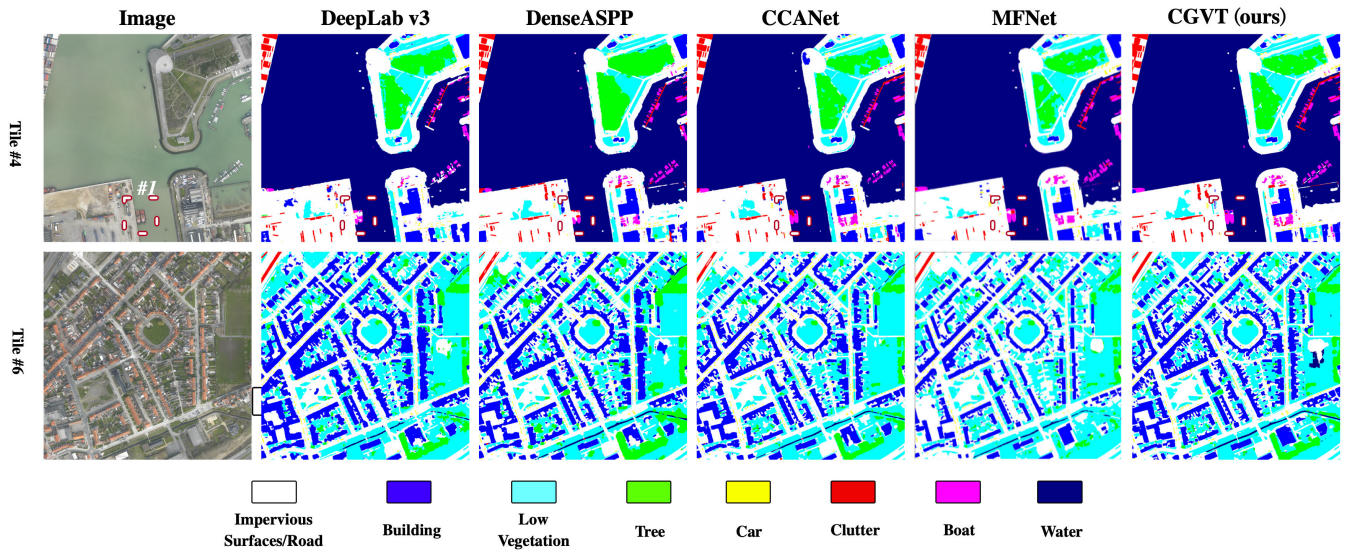


Fig. 8. Qualitative comparison between the proposed CGVT and published results on the Zeebrugge dataset.

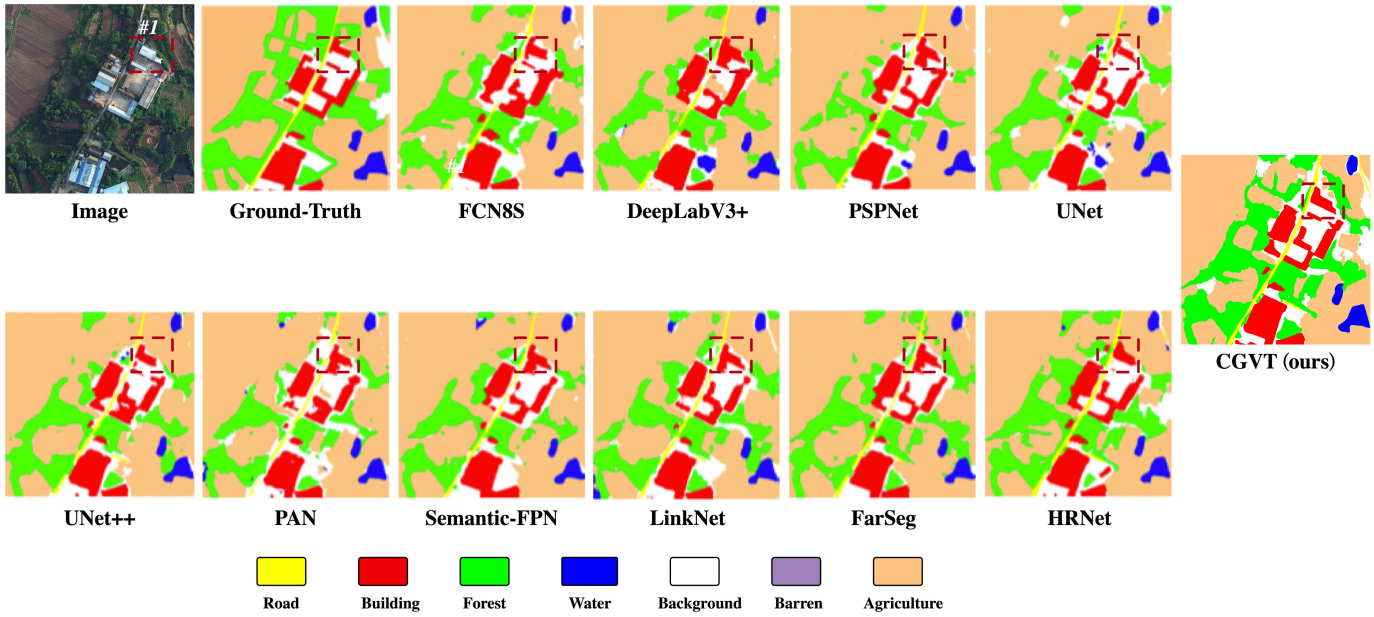


Fig. 9. Qualitative comparison between the proposed CGVT and published results for the LoveDA dataset.

the Zeebrugge dataset based on the results from published or preprint papers. In Fig. 8, it can be seen that the boundary of our segmentation is smoother and more accurate in terms of boundary details, for example, in box #1 of Fig. 8, our method successfully segmented the boat with better boundary detail.

3) Results on LoveDA Dataset: We submitted our results to the LoveDA Semantic Segmentation Challenge evaluation website to obtain the benchmark results for the test dataset. The same evaluation metric, mIoU, was used as the evaluation criterion on the evaluation website. The quantitative comparison results for the LoveDA dataset are shown in Table VII, including those reported in published or preprint papers. Up to February 28, 2022, we were in second place in the challenge competition. Compared with the results listed in the baseline [18], our mIoU is much higher than other methods, from 49.79% to 53.07%. In terms of the IoU of each class, our results are almost on par with the state-of-the-art performance, in contrast to the baseline [18], with large improvements in the following categories: buildings, water, barren, and agriculture. Compared to [65], our results represent improvements of 3.93%, 7.11%, 5.11%, and 2.19%, respectively.

The LoveDA dataset contains both rural and urban areas, and the geographical entities of these two areas have large differences. For example, urban areas contain more manual objects, whereas rural areas have more agricultural land, which leads to category imbalance and greatly improves the generalization of the model. In addition, buildings in rural areas are smaller than buildings in urban areas, and such scale differences can also increase the difficulty of segmentation; however, our method has a greater effect on the categories building, water, barren, and agriculture, which are more difficult to distinguish, indicating the strong generalization of our method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED CGVT AND OTHER PUBLISHED RESULTS ON THE LOVEDA DATASET. BOLDFACE IS THE BEST RESULT

Method	IoU per category (%)							mIoU (%)
	Background	Building	Road	Water	Barren	Forest	Agriculture	
FCN8S [19]	42.60	49.51	48.05	73.09	11.84	43.49	58.30	46.69
DeepLabV3+ [25]	42.97	50.88	52.02	74.36	10.40	44.21	58.53	47.62
PAN [66]	43.04	51.34	50.93	74.77	10.03	42.19	57.65	47.13
UNet [22]	43.06	52.74	52.78	73.08	10.33	43.05	59.87	47.84
UNet++ [67]	42.85	52.58	52.82	74.51	11.42	44.42	58.80	48.20
Semantic-FPN [44]	42.93	51.53	53.43	74.67	11.21	44.62	58.68	48.15
PSPNet [68]	44.4	52.13	53.52	76.50	9.73	44.07	57.85	48.31
LinkNet [69]	43.61	52.07	52.53	76.85	12.16	45.05	57.25	48.50
FarSeg [70]	43.09	51.48	53.85	76.61	9.78	43.33	58.90	48.15
FactSeg [71]	42.60	53.63	52.79	76.94	**16.20**	42.92	57.50	48.94
HRNet [65]	44.61	55.34	57.42	73.96	11.07	45.25	60.88	49.79
CGVT-Large (ours)	**46.97**	**59.27**	**57.52**	**81.07**	16.18	**47.39**	**63.07**	**53.07**

TABLE VIII
QUANTITATIVE COMPARISON BETWEEN DIFFERENT MODEL SCALES ON THE VAIHINGEN DATASET. ALL QUALITY MEASURES EXCEPT FOR OA (%) ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Methods	Impervious Surfaces/Road	Building	Low Vegetation	Tree	Car	OA
CGVT-S	92.0	95.2	82.6	89.2	87.5	91.2
CGVT-M	92.2	95.5	82.8	89.1	87.9	91.6
CGVT-L	**94.1**	**95.9**	**84.2**	**90.0**	**88.5**	**92.0**

From a qualitative perspective, Fig. 9 shows the visualization results of the methods listed in Table VII. It is obvious that our method has a better performance than the other methods. For example, FCN8S barely recognizes the road owing to the lack of feature extraction capability. Other methods also miss the segmentation of buildings, and the boundaries are not sufficiently clear, such as the red dashed box #1. In contrast, our method almost achieves a more perfect segmentation result where several methods do not perform well. As with the Zeebrugge dataset, all methods performed better for water bodies, which have strong spectral homogeneity and low intra-class variance. HRNet has a better performance than the other methods, but not as good as our method in terms of overall boundary detail and scattered trees.

B. Model Analysis

1) Different Model Scales and Decoders: As mentioned above, our model has three different scales, CGVT-S, CGVT-M, and CGVT- L, following the standard ViT model. The performances of these different model scales are shown in Tables VIII–X, respectively. As the results show, the accuracy on the three different datasets increases gradually with increasing model size and complexity. To intuitively observe the complexity of the models with different sizes, we analyzed the number of model parameters and giga floating-point operations per second (GFLOPS), with a test image size of 256 × 256, as shown in detail in Table XI.

For fair comparison, we used the same decoder (this table is UperNet head) for different backbones. In the table, we can see that the size of CGVT-S is basically equal to ResNet50 or even slightly lower than ResNet50, but our performance is still excellent. As mentioned earlier, our network uses different decoders, FPN decoder and UperNet decoder, and we tested both decoders separately on the Vaihingen dataset with CGVT-L as the backbone; the OA accuracy was 90.8%. The GFLOPs of the FPN decoder is only 27.29, which is only 1/3 of that in the UperNet decoder; however, the OA is only reduced by 1.2%, which fully proves that CGVT has good feature expression ability.

2) Effect of Pretraining on ImageNet: As our CGVT is a novel backbone, we did pretraining on the ImageNet 1K dataset to achieve better results for the subsequent segmentation. Our experimental results obtained with and without pretraining on the Vaihingen dataset are shown in Table XII. It can be seen that there is an abrupt fall in the values when there is no pretraining. This demonstrates that the pretraining process is important for applying transformer to vision tasks.

3) Ablation Study: The overall CGVT is composed of the crisscross-global transformer encoder. The crisscross-global transformer encoder is mainly composed of two different transformer encoder blocks: CC-TEB and GS-TEB. To evaluate the validity of each block, we combined the CC-TEB and GS-TEB independently. Specifically, we redesigned the original transformer encoder into two types, one with all GS-TEBs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IX

QUANTITATIVE COMPARISON BETWEEN DIFFERENT MODEL SCALES ON THE ZEEBRUGGE DATASET. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%), AA (%), AND KAPPA ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/ Road	Water	Clutter	Low Vegetation	Building	Tree	Boat	Car	OA	AA	Kappa
CGVT-S	90.05	98.88	65.26	78.80	86.72	80.68	69.24	90.82	88.91	82.99	0.85
CGVT-M	90.38	99.01	**69.09**	82.37	84.56	80.51	71.55	**91.55**	89.48	83.25	0.86
CGVT-L	**90.46**	**99.06**	66.87	**82.65**	**87.56**	**83.39**	**76.39**	91.34	**89.86**	**84.59**	**0.87**

TABLE X

QUANTITATIVE COMPARISON BETWEEN DIFFERENT MODEL SCALES ON THE LOVEDA DATASET. BOLDFACE IS THE BEST RESULT

Method	IoU per category (%)							mIoU (%)
	Background	Building	Road	Water	Barren	Forest	Agriculture	
CGVT-S	45.23	58.40	55.73	79.85	14.92	**47.68**	60.57	51.77
CGVT-M	45.33	58.64	56.32	80.01	15.23	47.25	61.65	52.06
CGVT-L	**46.97**	**59.27**	**57.52**	**81.07**	**16.18**	47.39	**63.07**	**53.07**

TABLE XI

COMPARISONS OF VARIANTS OF DIFFERENT MODEL DECODERS AND THEIR OA (%) ON VAIHINGEN DATASET. THE TEST DATA WERE 256 × 256 IMAGE PATCHES. BOLDFACE IS THE BEST RESULT

Backbone	Decoder					
	UperNet			FPN		
	Number of Parameters (Mega)	GFLOPS (Giga)	OA	Number of Parameters (Mega)	GFLOPS (Giga)	OA
ResNet50	66.42	59.25	86.21	-	-	-
ResNet101	85.41	64.11	88.35	-	-	-
CGVT-S	57.38	56.81	91.2	31.32	9.5	89.9
CGVT-M	95.64	65.14	91.6	67.59	17.71	90.3
CGVT-L	146.71	74.59	**92.0**	116.79	27.29	90.8

TABLE XII

RESULTS OBTAINED ON THE VAIHINGEN DATASET WITH/ WITHOUT PRETRAINING. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%) ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/Road	Building	Low Vegetation	Tree	Car	OA
CGVT-L (without pretrained)	92.6	93.8	80.5	87.2	82.4	88.4
CGVT-L (with pretraining)	**94.1**	**95.9**	**84.2**	**90.0**	**88.5**	**92.0**

TABLE XIII

ABLATION STUDY ON THE VAIHINGEN DATASET. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%) ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/Road	Building	Low Vegetation	Tree	Car	OA
All CG-TEBs	91.1	90.4	70.3	85.2	73.2	87.6
All CC-TEBs	**93.3**	**92.3**	**80.2**	**86.3**	**86.7**	**89.5**

and the other with all CC-TEBs. The results obtained on the Vaihingen dataset are shown in Table XIII. As can be seen, both structures lead to a certain degree of accuracy degradation. Specifically, in the first case, if the network is composed of all CC-TEBs, the accuracy degradation is obvious, because the global attention operation leads the network to represent local information, which is very unfavorable to the recognition of fine-grained objects in the VHR remote sensing images, for example, the F1 score of car is only 73.2%. In the second case, if the network is composed of all GS-TEBs, the accuracy also decreases to a certain extent, which is due to the limitation of the global information expression of the network. In summary, we account for the combined expression of global and local information by combining CC-TEB and GS-TEB.

In addition, we compared the two squeeze methods mentioned in GS-TEB, $AS_{pooling}$ and AS_{conv}, as shown in Table XIV. The results show that there is only marginal difference between the two methods; however, as AS_{conv} performs slightly better, the AS methods in this article are based on AS_{conv}.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
COMPARISONS OF DIFFERENT AS METHODS ON THE VAIHINGEN DATASET. ALL QUANTITATIVE MEASURES EXCEPT FOR OA (%) ARE F1 SCORES (%). BOLDFACE IS THE BEST RESULT

Method	Impervious Surfaces/Road	Building	Low Vegetation	Tree	Car	OA
$AS_{pooling}$	93.9	95.7	84.1	89.8	87.9	91.8
AS_{conv}	**94.1**	**95.9**	**84.2**	**90.0**	**88.5**	**92.0**

VI. CONCLUSION

We designed a novel end-to-end deep network called CGVT for VHR aerial image semantic segmentation. CGVT is a transformer-based architecture encoder backbone model that leverages the characteristics of the transformer to acquire long-range contextual information and successfully applies it to remote sensing image segmentation. To the best of our knowledge, this is the first high-resolution remote sensing image segmentation model that uses the new transformer architecture as the backbone without using CNNs. The model is mainly composed of two different transformer encoder blocks: CC-TEB and GS-TEB. We combine CC-TEB and GS-TEB in a reasonable manner to solve the problem of high computational effort and balance between local–global information acquisition in the application of the transformer to remote sensing image segmentation.

We validated the proposed model on three publicly available high-resolution remote sensing image segmentation datasets: Vaihingen, Zeebrugge, and LoveDA. Based only on raw image data without any postprocessing, our proposed CGVT achieved highly competitive performance for both quantitative and qualitative aspects. It achieved state-of-the-art performance on the Zeebrugge datasets and is ranked second in the semantic segmentation challenge on the LoveDA dataset and Vaihingen dataset.

ACKNOWLEDGMENT

The authors would like to thank the Belgian Royal Military Academy and the IEEE GRSS Image Analysis and Data Fusion Technical Committee for acquiring and providing the data used in this study. The numerical calculations in this article were performed on the supercomputing system in the Supercomputing Center, Wuhan University, Wuhan, China.

REFERENCES

[1] L. Yishao, B. Jiajun, C. Chun, and S. Mingli, "A novel image segmentation algorithm based on convex polygon edge detection," in *Proc. IEEE Region 10 Conf. (TENCON)*, 2004, pp. 108–111.

[2] N. Ikonomakis, K. N. Pataniotis, and A. N. Venetsanopoulos, "Unsupervised seed determination for a region-based color image segmentation scheme," in *Proc. Int. Conf. Image Process.*, 2000, pp. 537–540.

[3] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. Dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.

[4] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 78–95, Nov. 2018.

[5] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[6] Y. Yuan, L. Huang, J. Guo, C. Zhang, X. Chen, and J. Wang, "OCNet: Object context for semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 8, pp. 2375–2398, Aug. 2021.

[7] M. Xia, T. Wang, Y. Zhang, J. Liu, and Y. Xu, "Cloud/shadow segmentation based on global attention feature fusion residual network for remote sensing imagery," *Int. J. Remote Sens.*, vol. 42, no. 6, pp. 2022–2045, Mar. 2021.

[8] J. Yang, W. An, C. Yan, P. Zhao, and J. Huang, "Context-aware domain adaptation in semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 514–524.

[9] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Object detectors emerge in deep scene CNNs," 2014, *arXiv:1412.6856*.

[10] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[11] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–12.

[12] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[13] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[14] X. Chu et al., "Twins: Revisiting the design of spatial attention in vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 9355–9366.

[15] X. Dong et al., "CSWin transformer: A general vision transformer backbone with cross-shaped windows," 2021, *arXiv:2107.00652*.

[16] ISPRS. *International Society for Photogrammetry and Remote Sensing 2D Semantic Labeling Challenge*. Accessed: Feb. 21, 2022. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/semanticlabeling.html

[17] M. Campos-Taberner et al., "Processing of extremely high-resolution LiDAR and RGB data: Outcome of the 2015 IEEE GRSS data fusion contest—Part A: 2-D contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5547–5559, Aug. 2016.

[18] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[19] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[20] W. Sun and R. Wang, "Fully convolutional networks for semantic segmentation of very high resolution remotely sensed images combined with DSM," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 474–478, Mar. 2018.

[21] L. Mou and X. X. Zhu, "RiFCN: Recurrent network in fully convolutional network for semantic segmentation of high resolution remote sensing images," 2018, *arXiv:1805.02091*.

[22] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[23] L. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[24] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[25] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

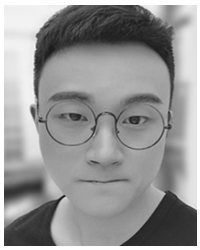
[26] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[27] X. Zheng, X. Wu, L. Huan, W. He, and H. Zhang, "A gather-to-guide network for remote sensing semantic segmentation of RGB and auxiliary image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[28] Y. Sun, Z. Fu, C. Sun, Y. Hu, and S. Zhang, "Deep multimodal fusion network for semantic segmentation using remote sensing image and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.

[29] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, Mar. 2020.
[30] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.
[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[32] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.
[33] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," 2020, *arXiv:2012.12877*.
[34] Y. Wang et al., "End-to-end video instance segmentation with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8741–8750.
[35] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.
[36] L. Gao et al., "STransFuse: Fusing Swin transformer and convolutional neural network for remote sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10990–11003, 2021.
[37] Z. Xu, W. Zhang, T. Zhang, Z. Yang, and J. Li, "Efficient transformer for remote sensing image segmentation," *Remote Sens.*, vol. 13, no. 18, p. 3585, Sep. 2021.
[38] P. Shaw, J. Uszkoreit, and A. Vaswani, "Self-attention with relative position representations," 2018, *arXiv:1803.02155*.
[39] X. Chu, Z. Tian, B. Zhang, X. Wang, and C. Shen, "Conditional positional encodings for vision transformers," 2021, *arXiv:2102.10882*.
[40] M. Amirul Islam, S. Jia, and N. D. B. Bruce, "How much position information do convolutional neural networks encode?" 2020, *arXiv:2001.08248*.
[41] A. Vaswani, P. Ramachandran, A. Srinivas, N. Parmar, B. Hechtman, and J. Shlens, "Scaling local self-attention for parameter efficient visual backbones," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12894–12904.
[42] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[43] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.
[44] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6399–6408.
[45] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[46] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
[47] *International Society for Photogrammetry and Remote Sensing 2D Semantic Labeling Challenge*. Accessed: Mar. 10, 2022. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html
[48] (2015). *IEEE GRSS Data Fusion Contest*. [Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion
[49] A. Paszke et al., "Automatic differentiation in PyTorch," in *Proc. NIPS*, 2017, pp. 1–4.
[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[51] A. Garcia-Garcia, S. Orts-Escolano, S. Oprea, V. Villena-Martinez, and J. Garcia-Rodriguez, "A review on deep learning techniques applied to semantic segmentation," 2017, *arXiv:1704.06857*.
[52] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.
[53] H. Bai, J. Cheng, X. Huang, S. Liu, and C. Deng, "HCANet: A hierarchical context aggregation network for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[54] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2017.
[55] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.
[56] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-Net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[57] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2021.
[58] S. Piramanayagam, E. Saber, W. Schwartzkopf, and F. Koehler, "Supervised classification of multisensor remotely sensed images using a deep learning framework," *Remote Sens.*, vol. 10, no. 9, p. 1429, Sep. 2018.
[59] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[60] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.
[61] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.
[62] N. Audebert, A. Boulch, B. Le Saux, and S. Lefèvre, "Distance transform regression for spatially-aware deep semantic segmentation," *Comput. Vis. Image Understand.*, vol. 189, Dec. 2019, Art. no. 102809.
[63] Y. Liu, S. Piramanayagam, S. T. Monteiro, and E. Saber, "Semantic segmentation of multisensor remote sensing imagery with deep ConvNets and higher-order conditional random fields," *J. Appl. Remote Sens.*, vol. 13, no. 1, p. 016501, 2019.
[64] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.
[65] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," 2018, *arXiv:1805.10180*.
[66] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20, 2018, Proceedings 4*. Springer, 2018, pp. 3–11.
[67] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4.
[68] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.
[69] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.



Guohui Deng received the B.S. degree in geographic information systems from Anhui Normal University, Wuhu, China, in 2016, where he is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the School of Remote Sensing and Information Engineering.

His research interests include high-resolution satellite image classification, machine learning, and deep learning.



Zhaocong Wu received the B.S. and Ph.D. degrees in photogrammetry and remote sensing techniques from Wuhan University, Wuhan, China, in 1986 and 2004, respectively.

He is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests include high-resolution image processing, image analysis, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.



Miaozhong Xu received the B.S. degree from the Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1989, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, in 2004.

He is currently a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing. His research interests include high-resolution image processing, photogrammetry, and cartography.



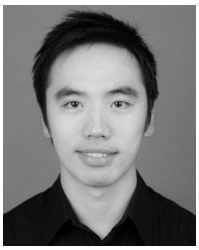
Chengjun Wang received the B.S. degree in surveying and mapping from Fuzhou University, Fuzhou, China, in 2018, where he is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing.

His research interests include scene classification of high-resolution satellite images, deep learning, and computation vision in remote sensing applications.



Zhiye Wang received the bachelor's degree in remote sensing from Wuhan University, Wuhan, China, in 2014, the master's degree in geodesy and geo-information from the Technical University of Munich (TUM), Munich, Germany, in 2019. He is currently pursuing the Ph.D. degree with LIESMARS, Wuhan University.

His research interests include remote sensing, computer vision, and application in power infrastructure detection and surveillance



Zhongyuan Lu received the B.S. degree in geomatics engineering from University of Stuttgart, Stuttgart, Germany, in 2010. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include satellite image classification, deep learning, and computer vision applications in remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:32:08 UTC from IEEE Xplore. Restrictions apply.